BATCH 18

# Reduced-Order Simulation of an Entrained-Flow Coal Gasifier

BYU atmospheric reactor | Utah bituminous coal | Mularski and Modlinski (2021)

The work uses the published BYU test conditions and compares a transparent Python model with reported experimental and CFD results. The numerical predictions are presented with their limits, including an unresolved temperature ceiling and a substantial conversion gap.

## Group roster

| Roll number | Student |
|---|---|
| 23ME31035 | Upendra Singh |
| 23ME31025 | Samrudh Nelli |
| 23ME10102 | Arnav Chauhan |
| 23ME3AI34 | Ashutosh Ranjan |
| 23MF10027 | Naveen Saini |

# Abstract

This project models the BYU atmospheric entrained-flow gasifier using the Utah bituminous coal and Table 3 streams selected in Mularski and Modlinski (2021). One reduced-model workflow applies axial reacting-lane chemistry across radial annuli, with radial species and heat exchange. The 1-D and radial descriptions refer to spatial detail within this workflow, not separate project phases. The model does not solve momentum, particle trajectories, radiation or recirculation. At 120 radial by 600 axial stations, elemental balances close within 0.000003%, while the outlet-area-average temperature is 3556.7 K and the recovered peak reaches the 4000 K solver ceiling. The predicted carbon conversion is 53.52%, compared with 82% in the BYU experiment; dry inert-free CO, H2, CO2 and CH4 are 58.66, 16.68, 24.21 and 0.45 vol%, respectively. The model also leaves 3.88 vol% dry O2 at the outlet. Grid refinement changes the outlet temperature by 9.1 K and conversion by 1.28 percentage points from 80x400 to 120x600. These checks support numerical bookkeeping, but the outlet disagreement and ceiling prevent a claim of physical validation. A mass-weighted Rosin-Rammler sweep using the cross-section-averaged calculation exposes saturation in the current chemistry.

## Study scope and outputs

| Analysis area | Evidence presented |
|---|---|
| Feed composition | Utah coal proximate and DAF ultimate analyses; primary and secondary streams. |
| Species distributions | Cross-sectional area-average axial profiles for CO, H2, CO2, H2O, O2 and CH4. |
| Species contours | Radial-axial contours for CO, H2, CO2, H2O and O2. |
| Reactor temperature | Temperature contour and area-average axial temperature profile. |
| Comparison with journal results | Outlet composition/conversion comparison plus a near-axis profile comparison from the radial calculation against digitized experimental markers from Mularski et al. (2021), Figure 9. |
| Conclusions | Quantitative findings, model limits and validation gaps. |

# 1. Selected case and feed

The case is the BYU atmospheric entrained-flow reactor reported by Mularski and Modlinski (2021): 0.20 m internal diameter, 1.80 m length, 1 atm pressure, and a 23.88 kg/h Utah bituminous coal feed.

## Coal analysis

| Analysis | Basis | Component | wt% |
|---|---|---|---|
| Proximate | As received | Moisture | 2.40 |
| Proximate | As received | Volatile matter | 45.60 |
| Proximate | As received | Fixed carbon | 43.70 |
| Proximate | As received | Ash | 8.30 |
| Ultimate | Dry ash free | Carbon | 77.60 |
| Ultimate | Dry ash free | Hydrogen | 6.56 |
| Ultimate | Dry ash free | Nitrogen | 1.42 |
| Ultimate | Dry ash free | Sulfur | 0.55 |
| Ultimate | Dry ash free | Oxygen | 13.87* |

* Oxygen is calculated by difference from the reported DAF C, H, N and S fractions. The HHV used in the code is estimated as 33.52 MJ/kg DAF from the Channiwala-Parikh correlation because a measured HHV was not entered for this case.

## Operating streams

| Stream | Mass flow | Inlet T | Mole composition |
|---|---|---|---|
| Primary | 26.24 kg/h | 400 K | 85.0% O2, 12.6% Ar, 2.4% H2O; coal carried in this stream |
| Secondary | 6.62 kg/h | 473 K | 100% H2O |
| Coal | 23.88 kg/h | - | Utah bituminous coal; 1-85 um Rosin-Rammler distribution |

The 2021 paper specifies a Rosin-Rammler scale diameter of 36 um and spread parameter 1.033. The radial inlet profile also uses 13 mm primary-nozzle ID and 28.6 mm secondary-annulus OD from Abani and Ghoniem (2013), because the 2021 case table does not list nozzle sizes. Density (1300 kg/m3), 20% background inlet spreading and the radial mixing coefficient are explicit assumptions, not measurements.

# 2. Model formulation

The implementation assumes steady, ideal-gas flow at constant pressure. Its cross-section-averaged 1-D calculation uses plug flow, one gas-particle temperature and instantaneous moisture release. Temperature is recovered from an adiabatic enthalpy balance. An explicit 1400 K inlet ignition patch supplies the hot start that a cold plug-flow model cannot generate without recirculation. Summed across the 120 radial annuli, that patch adds 19.35 kW, or 9.75% of the coal HHV input rate. It stands in for flame stabilization and recirculated heat; it is a material model assumption, not a negligible numerical detail.

The radial calculation allocates primary gas, secondary steam and coal to annular stream tubes using the nozzle areas. It applies the same axial chemistry in each lane and couples neighboring lanes through radial transport. Species exchange is solved with backward-Euler finite volumes. Species enthalpy is transferred with the gas, followed by a conservative radial heat-conduction step. The turbulent diffusivity follows $D_t = C_{mix} U_{bulk} R$ with $C_{mix} = 0.05$, turbulent Schmidt number 0.70 and turbulent Prandtl number 0.90. A molecular floor and a 0.20 m2/s cap are imposed. The wall remains adiabatic. These 1-D and radial calculations are spatial components of one model workflow.

The C2SM devolatilization constants are those reported in Table 5 of Mularski and Modlinski (2021), fitted there against FG-DVC. Char oxidation and gasification use that paper's BYU-fitted kinetic-diffusion parameters from Table 6 and its film-diffusion coefficient from Equation (3). The project does not rerun the FG-DVC or CBK/E-CBK/G fitting workflow. Gas chemistry is reduced to four global reactions and does not use the paper's full turbulence-chemistry coupling. The volatile species split adds assumptions to close the elemental balance; surplus carbon goes to a soot/char pool rather than explicit tar. Thus the published operating point and selected chemistry parameterizations are used, while the complete CFD model is not reproduced.

## Difference from the selected CFD

| Feature | Mularski and Modlinski (2021) | Batch 18 implementation |
|---|---|---|
| Flow field | 2-D axisymmetric RANS; realizable k-epsilon turbulence | Radial stream tubes; no momentum equation |
| Particle transport | Lagrangian particles with stochastic dispersion | Particles remain in their inlet annulus |
| Radiation | Discrete ordinates with WSGG gas radiation | Adiabatic wall; no radiation |
| Chemistry | C2SM/FG-DVC devolatilization, fitted kinetic-diffusion char model, and EDC global gas reactions | Published C2SM and BYU-fitted char parameters; four global gas reactions without EDC |
| Tar and mixing | Explicit tar, turbulent mixing and turbulence-chemistry coupling | Surplus volatile carbon becomes soot; operator-split radial diffusion |

The radial code is a reduced-order model for spatial trends and conservation checks. It does not reproduce the reference CFD equations or the full burner recirculation zone.

# 3. Numerical verification

Global C, H, O, N and Ar flows are checked at the outlet, counting gases, char and unreacted coal. On the 120x600 grid, the maximum absolute elemental imbalance is 0.0000021%. This demonstrates that the implemented source terms and radial transfers conserve the tracked elements. It does not establish that the rate laws or closure assumptions match the physical reactor.

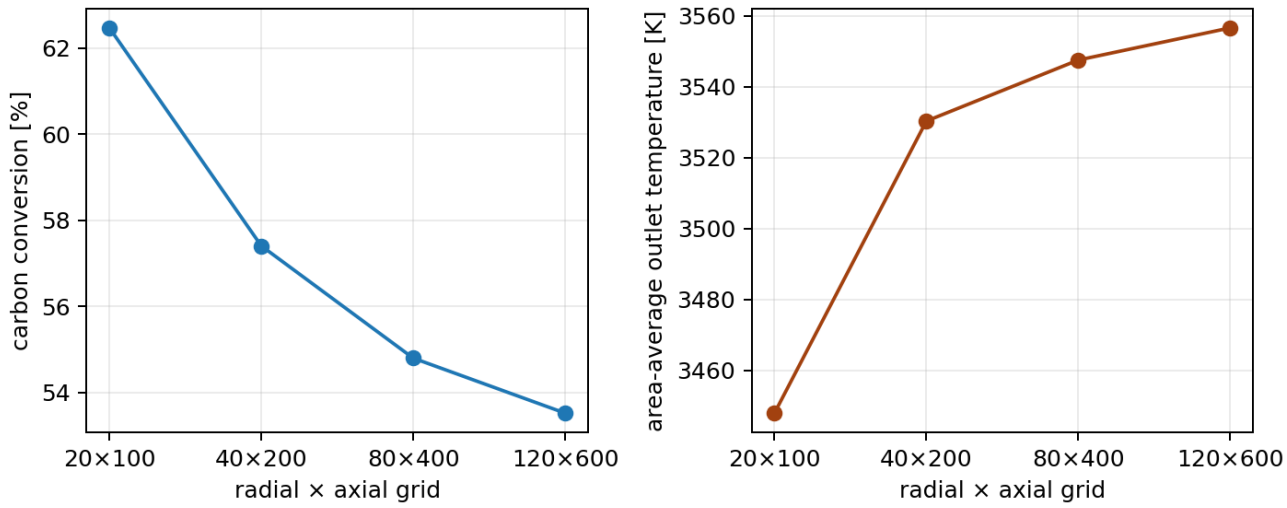



Figure 1. Grid refinement for carbon conversion and area-average outlet temperature. The peak temperature is 4000 K on all four grids because it reaches the recovery ceiling, so peak-temperature convergence cannot be assessed.

| Grid | Outlet T (K) | Conversion (%) | Dry CO (%) | Dry H2 (%) | Dry CO2 (%) |
|---|---|---|---|---|---|
| 20 x 100 | 3447.9 | 62.47 | 55.95 | 17.25 | 14.62 |
| 40 x 200 | 3530.3 | 57.41 | 53.46 | 14.87 | 18.73 |
| 80 x 400 | 3547.6 | 54.80 | 50.71 | 14.98 | 19.86 |
| 120 x 600 | 3556.7 | 53.52 | 50.07 | 14.24 | 20.66 |

The grid-study species fractions are normalized to total dry outlet gas, including Ar, N2 and residual O2. The outlet comparison later uses a dry inert-free basis. From 80x400 to 120x600, outlet temperature shifts by 9.1 K, conversion falls by 1.28 percentage points, and the principal dry species shift by 0.65-0.80 points. This refinement trend does not establish physical accuracy; a separate analytical pure-diffusion test has not been completed.

# 4. Temperature and CO fields

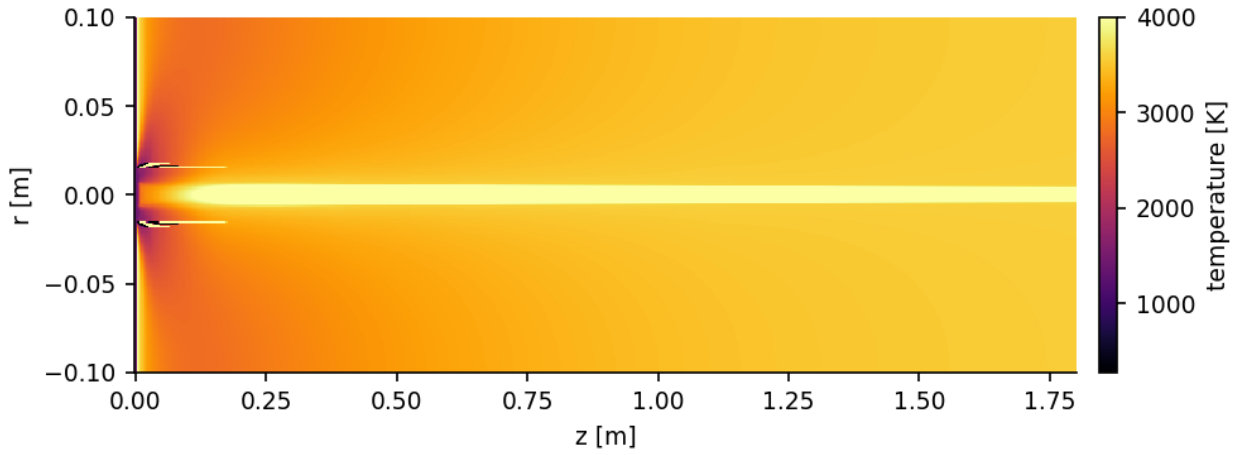



Figure 2. Temperature field (K), mirrored about the axis. The reported maximum is clipped at the 4000 K temperature-recovery ceiling; its true peak is unresolved.

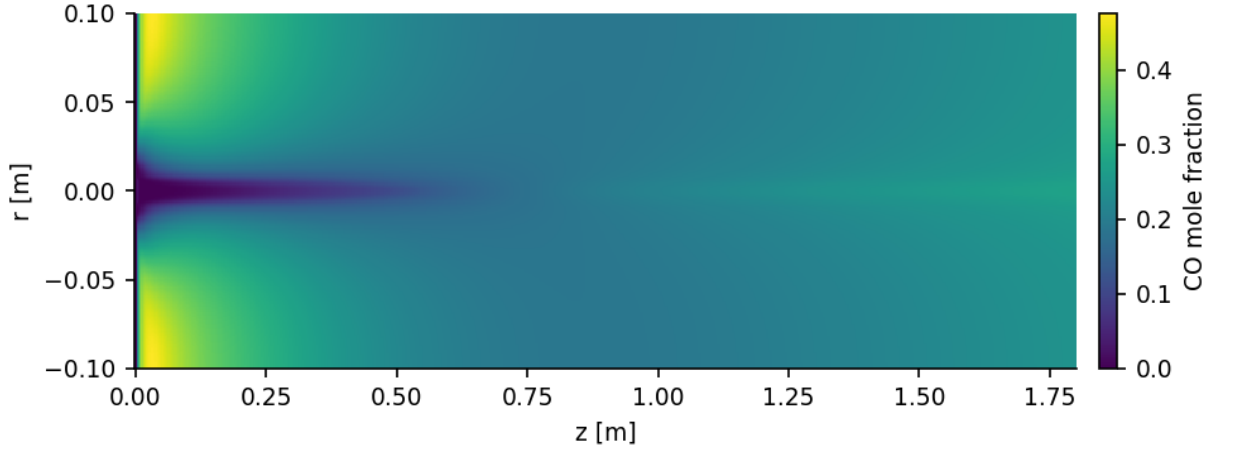



Figure 3. CO mole fraction. The annular model resolves radial redistribution by the assumed turbulent-diffusion closure; it has no radial momentum or particle-dispersion solution.

# 5. Hydrogen and carbon-dioxide fields

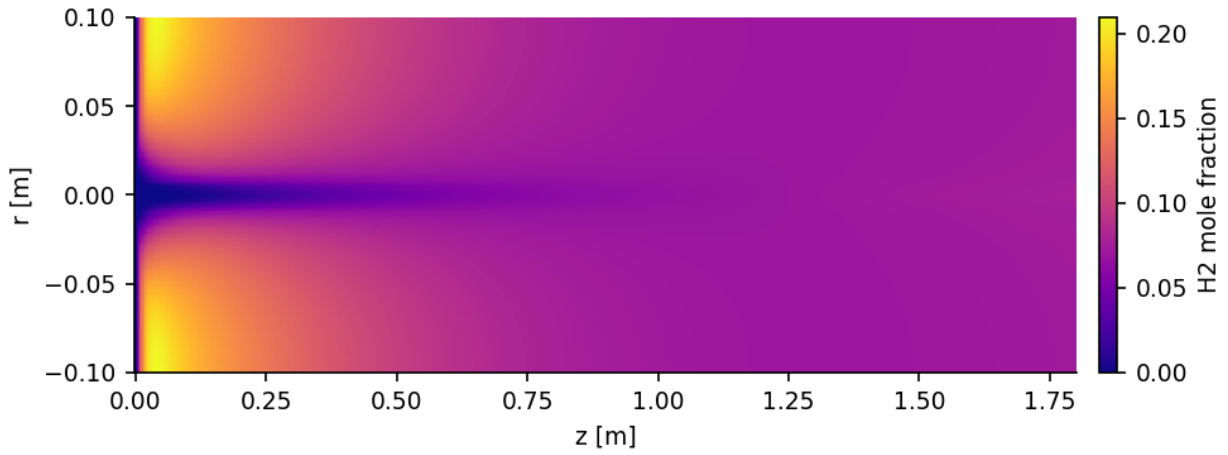



Figure 4. H2 mole fraction. In this model, hydrogen comes from the imposed volatile split, water-gas shift and char-steam reaction.

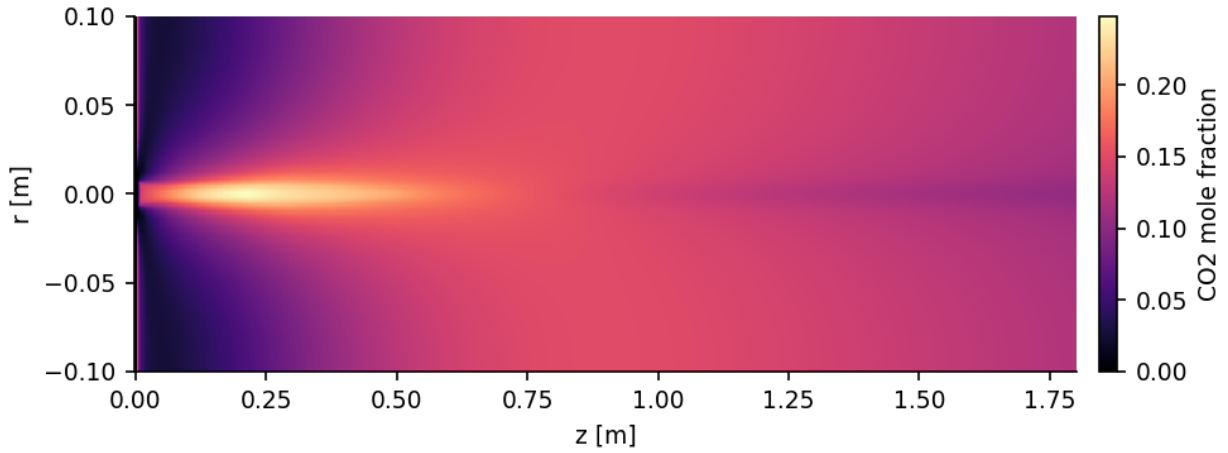



Figure 5. CO2 mole fraction. The published outlet comparison is more meaningful than apparent contour smoothness because the radial mixing coefficient is assumed.

# 6. Steam and methane fields

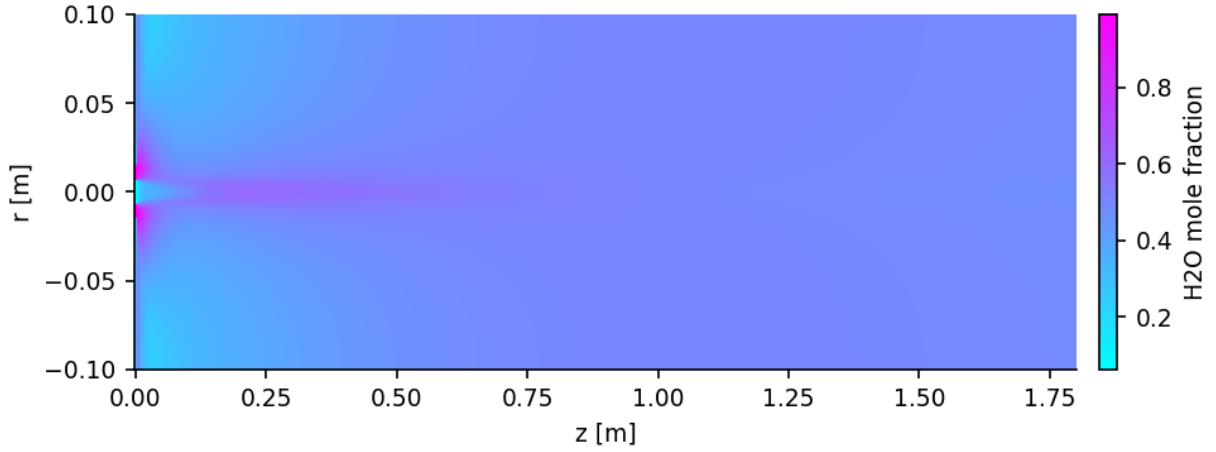



Figure 6. H2O mole fraction. The reference paper reports about +/-14% experimental uncertainty for water and excludes it from its model-quality score.

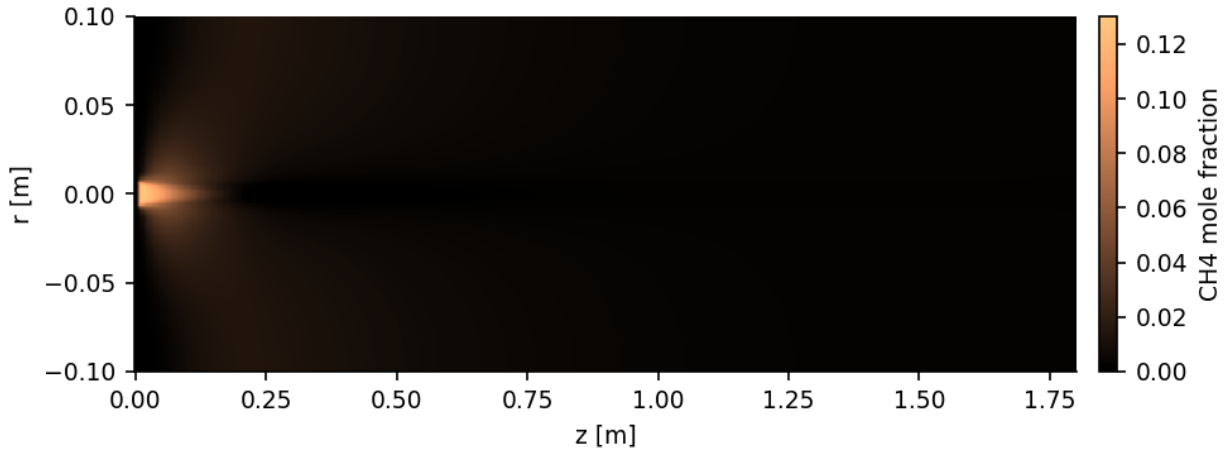



Figure 7. CH4 mole fraction. Methane is a minor outlet product in both the computed and measured dry inert-free streams.

# 7. Oxygen and carbon conversion fields

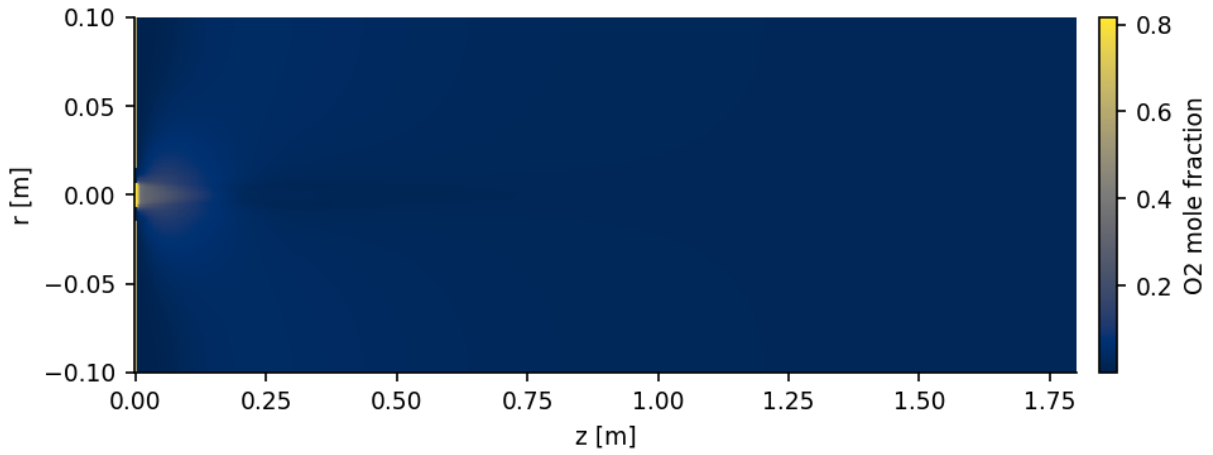



Figure 8. O2 mole fraction. The model retains 3.88 vol% O2 in its dry outlet, which flags incomplete oxygen consumption and a chemistry/mixing mismatch.

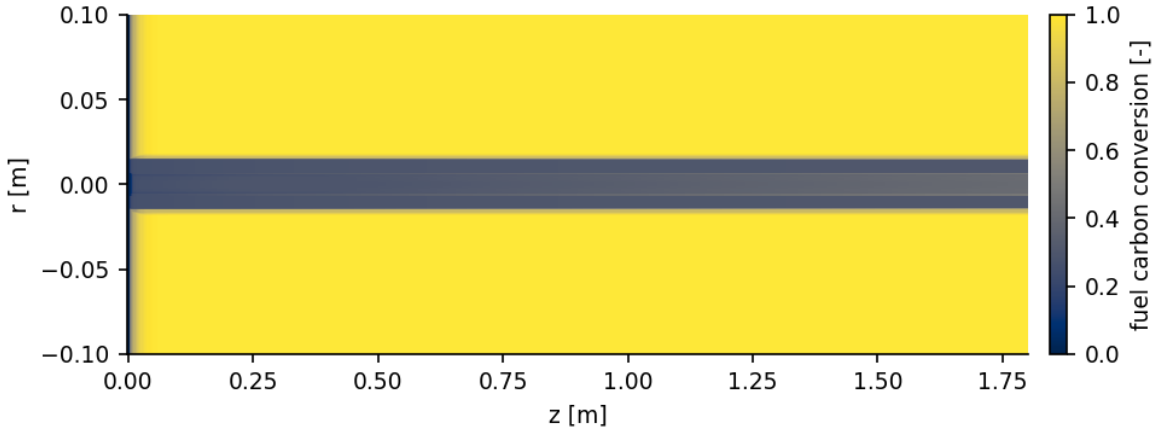



Figure 9. Local fuel-carbon conversion, calculated from remaining char and raw coal in each annulus.

# 8. Axial species and temperature profiles

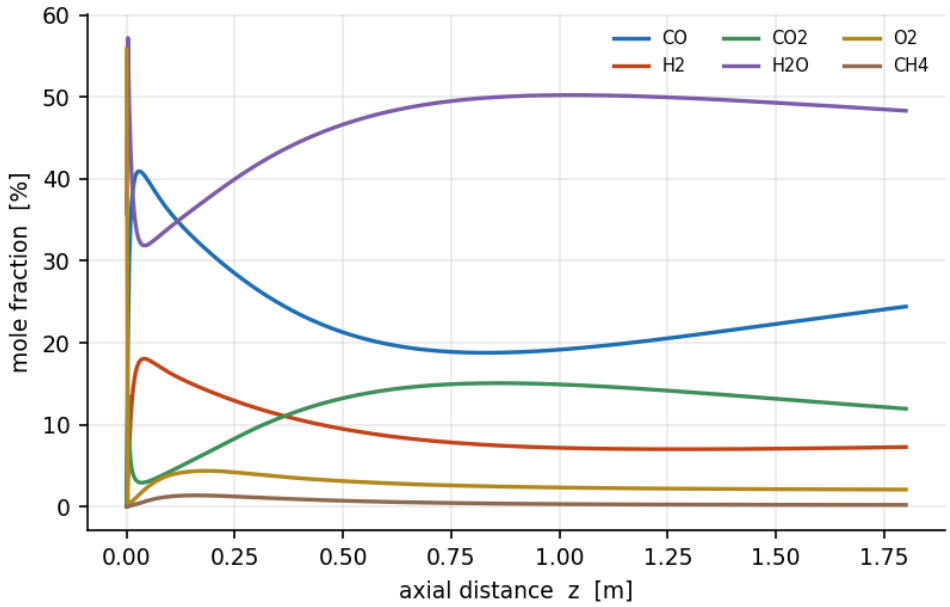



Figure 10. Cross-sectional area-average species profiles. The pointwise experiment comparison uses the near-axis model profile separately, because the reference measurements are centerline data.

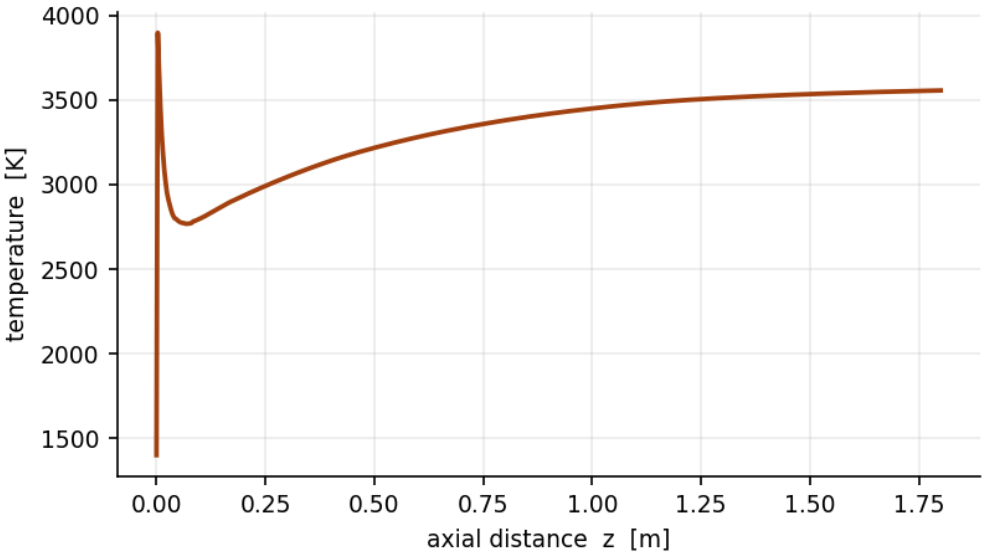



Figure 11. Area-average temperature along the reactor. The outlet is 3556.7 K. The local peak reaches the numerical ceiling of 4000 K and must not be interpreted as a resolved physical temperature.

The selected paper describes a flame peak near 3000 K. The excessive and unresolved peak is consistent with the model's adiabatic wall, missing radiation, ignition patch equal to 9.75% of the coal HHV input rate, one-temperature treatment, and uncalibrated reduced reaction/mixing closures. The calculation is useful for locating where its own species fields change; the temperature field is not quantitatively validated.

| Radial-model result | Value | Interpretation |
|---|---|---|
| Peak temperature | 4000 K | At solver ceiling; unresolved |
| Outlet area-average T | 3556.7 K | Reduced-model output |
| Carbon conversion | 53.52% | 28.48 percentage points below experiment |
| Dry outlet O2 | 3.88 vol% | Residual oxygen indicates chemistry mismatch |

# 9. Near-axis comparison with centerline measurements

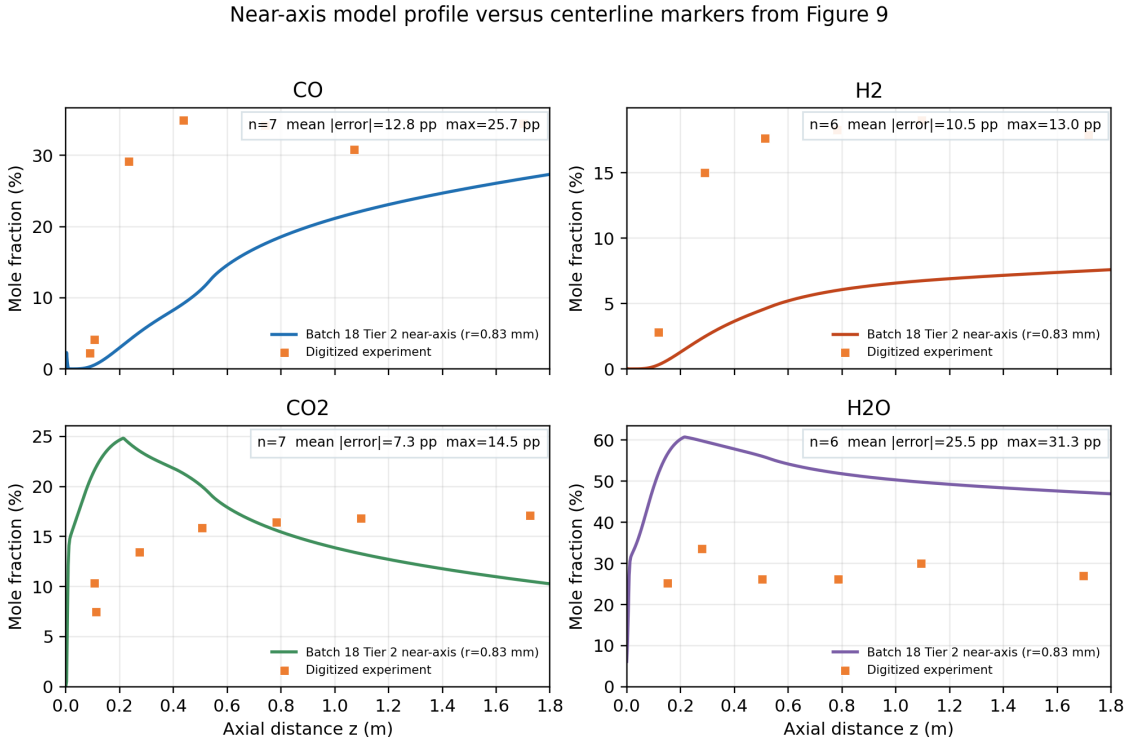



Figure 12. Near-axis profile from the radial calculation (innermost annulus center, r = 0.00083 m) compared with centerline experimental markers digitized from Figure 9 of Mularski and Modlinski (2021). Marker coordinates are approximate, with estimated digitization uncertainty of about +/-0.3 percentage point and +/-0.01 m.

| Species | n | Batch 18 mean / max (pp) | Paper Table 8 mean / max (pp) |
|---|---|---|---|
| CO | 7 | 12.81 / 25.72 | 4.82 / 18.48 |
| H2 | 6 | 10.50 / 13.03 | 5.47 / 7.95 |
| CO2 | 7 | 7.27 / 14.46 | 4.72 / 10.15 |
| H2O | 6 | 25.54 / 31.34 | 9.58 / 20.22 |

Errors are mean / maximum absolute differences in mole fraction, reported as percentage points at the digitized experimental axial locations. This is the same absolute-difference convention as the paper's Table 8, not a relative percentage error. H2O is shown for completeness but excluded from the paper's quality score because its experimental estimate has about +/-14% uncertainty. The large CO, H2 and CO2 gaps show that the current reduced model does not reproduce the measured axial profiles.

# 10. Radial profiles against experimental probes

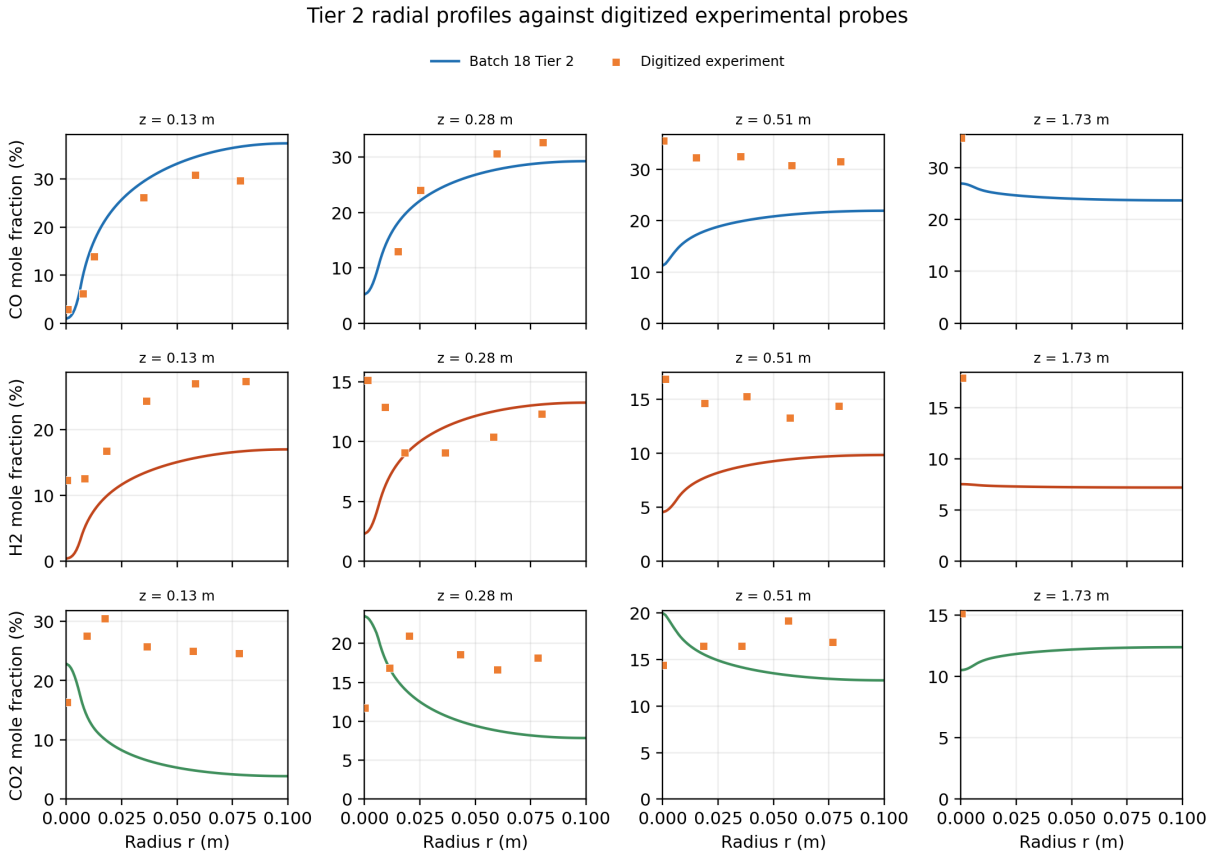



Figure 13. Radial-model profiles at four representative axial stations compared with orange experimental markers digitized from Figures 12-14 of Mularski and Modlinski (2021). All eight source stations were scored; the plot shows 0.13, 0.28, 0.51 and 1.73 m. Marker coordinates are approximate (estimated +/-0.001 m in radius and +/-0.5 percentage point in composition).

| Species | n | Mean / max absolute error (pp) |
|---|---|---|
| CO | 34 | 8.54 / 24.22 |
| H2 | 36 | 7.02 / 15.33 |
| CO2 | 35 | 6.74 / 20.51 |

The pooled scores cover 105 CO, H2 and CO2 probe points across eight axial stations. H2O is omitted because the paper derives it by hydrogen balance rather than reporting an independent measured profile. These comparisons use radial-model values interpolated to the digitized probe radii. The largest local discrepancies remain substantial, so the reduced model does not reproduce all measured radial structure.

# 11. Outlet comparison with published results

Brown et al. (1988) report a dry inert-free Utah-coal effluent composition and 82% carbon conversion. The model composition below is calculated from the summed outlet molar flow of the 120x600 radial calculation after removing H2O, Ar, N2 and O2, then renormalizing CO, H2, CO2 and CH4 to 100%. Differences are model minus reported experiment in percentage points.

| Quantity | Batch 18 model | Experiment | Difference |
|---|---|---|---|
| CO (vol%) | 58.66 | 51.1 | +7.56 pp |
| H2 (vol%) | 16.68 | 24.7 | -8.02 pp |
| CO2 (vol%) | 24.21 | 23.6 | +0.61 pp |
| CH4 (vol%) | 0.45 | 0.3 | +0.15 pp |
| Carbon conversion | 53.52% | 82% | -28.48 pp |

The conversion mismatch and residual outlet O2 show that elemental conservation alone is not validation. The model overpredicts CO, underpredicts H2 and converts too little solid carbon. Cold-gas efficiency is 25.95% on the LHV basis used here; it is not compared directly with the experimental value because the calorific-value basis and product accounting differ.

The paper's Table 8 values describe its own modified CFD parameters; they are shown on the preceding page as a reference benchmark, not attributed to Batch 18. The paper also reports carbon conversion of 83% for the 36 um monodisperse case and 77% for the 1-85 um size distribution, against 82% measured. These values check the paper's particle-size result, not a prediction reproduced by the reduced model.

# 12. Rosin-Rammler particle-size diagnostic

The paper's particle size distribution is represented as F(d) = 1 - exp[-(d/d_scale)^n], truncated to 1-85 um. Six diameter bins with equal increments in cumulative mass are run independently; outlet gas flows and fuel-based metrics are weighted by each bin's mass fraction. The study varies d_scale = 20, 36 and 60 um and n = 0.7, 1.033 and 1.5. A monodisperse 36 um run is also included.

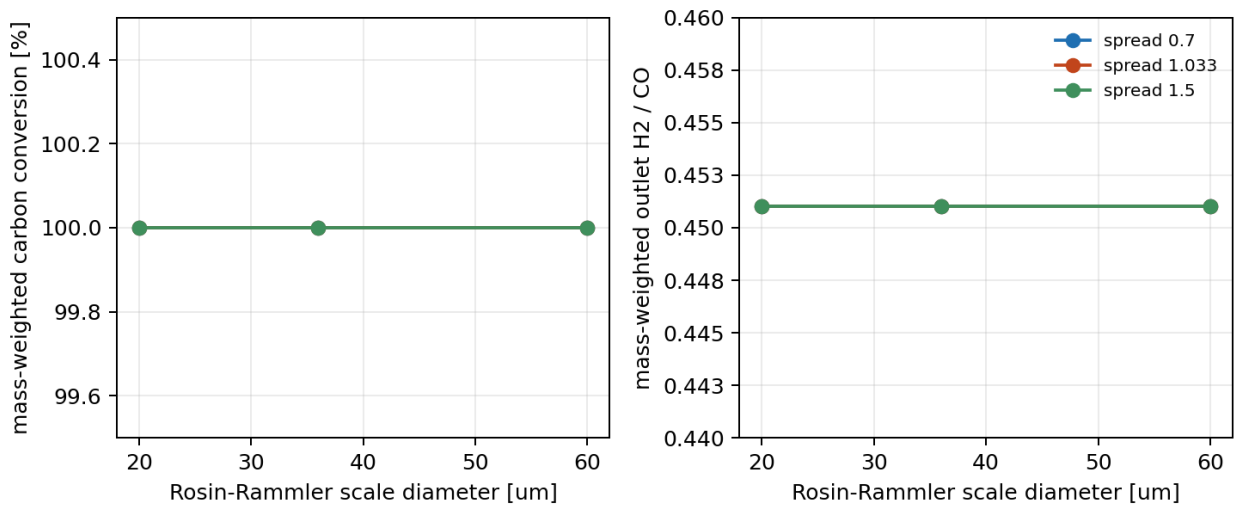



Figure 14. Mass-weighted carbon conversion and H2/CO response from the cross-section-averaged calculation. Each curve is flat because that calculation converts 100% of fuel carbon and returns identical outlet chemistry for every tested particle distribution.

| Distribution | Conversion | CGE | H2/CO |
|---|---|---|---|
| RR 36 um, n=1.033 | 100.00% | 71.65% | 0.451 |
| All 10 cases | 100.00% | 71.65% | 0.451 |
| Mularski 36 um mono | 83% | Reference CFD | Not listed here |
| Mularski full 1-85 um RR | 77% | Reference CFD | Not listed here |
| Brown et al. experiment | 82% | Experimental | Not listed here |

The flat result does not establish particle-size independence. It diagnoses a limitation in the cross-section-averaged calculation: complete burnout is reached for all distributions, leaving no resolved sensitivity. The extension is therefore documented as a failed predictive test rather than evidence that size does not matter.

# 13. Conclusions and reproducibility

1. The radial finite-volume transport and reaction bookkeeping conserve C, H, O, N and Ar to within 0.000003% on the finest grid.

2. The 80x400 to 120x600 refinement changes outlet temperature by 9.1 K, carbon conversion by 1.28 percentage points, and the principal dry species by less than 0.81 percentage points. Peak-temperature convergence remains unknown because every grid reaches the 4000 K ceiling.

3. The model predicts 53.52% carbon conversion and dry inert-free CO/H2/CO2/CH4 of 58.66/16.68/24.21/0.45 vol%, compared with 82% conversion and 51.1/24.7/23.6/0.3 vol% in the BYU experiment. It also predicts 3.88 vol% dry outlet O2.

4. The present model is numerically conservative but physically underpredicts carbon conversion, underpredicts H2 and leaves oxygen at the outlet. Near-axis mean absolute errors are 12.81 pp for CO, 10.50 pp for H2 and 7.27 pp for CO2. Across radial probes, pooled mean errors are 8.54 pp for CO, 7.02 pp for H2 and 6.74 pp for CO2. Its 1400 K hot-start patch adds 19.35 kW, equal to 9.75% of coal HHV input. It should be treated as a reduced model with unresolved peak temperature, not a validated replacement for the reference CFD.

5. The mass-weighted particle-size sweep is flat because the cross-section-averaged calculation saturates at 100% conversion. A credible size-effect result requires first fixing the chemistry and resolving finite char burnout.

## Run instructions

From the Batch18 folder: python scripts/run_tier1.py; python scripts/run_tier2.py; python scripts/run_grid_study.py; python scripts/sweep_particle_size.py; python scripts/digitize_radial_markers.py; python scripts/make_validation_plots.py; python scripts/make_radial_validation_plots.py. The detailed inputs are in case/byu_case.yaml. The radial-model run writes contours, area-average, near-axis and radial profiles, tier2_summary.csv and tier2_metrics.csv under report/. The validation scripts write the axial and radial comparison figures and error tables. The source code and run instructions are included in the Batch 18 source package.

# References

[1] Mularski, J. and Modlinski, N. (2021). Entrained-Flow Coal Gasification Process Simulation with the Emphasis on Empirical Char Conversion Models Optimization Procedure. Energies, 14(6), 1729. https://doi.org/10.3390/en14061729

[2] Brown, B. W., Smoot, L. D., Smith, P. J. and Hedman, P. O. (1988). Measurement and Prediction of Entrained-Flow Gasification Processes. AIChE Journal, 34(3), 435-446. Original article PDF: https://electronicsandbooks.com/edt/manual/Magazine/A/AIChE%20Journal%20US/1988%20v.34/03%20%28353-526%29/435-446.pdf

[3] Abani, N. and Ghoniem, A. F. (2013). Large eddy simulations of coal gasification in an entrained flow gasifier. Fuel, 104, 664-680. https://doi.org/10.1016/j.fuel.2012.06.006. Used only for the 13 mm primary and 28.6 mm secondary nozzle dimensions.

[4] Channiwala, S. A. and Parikh, P. P. (2002). A unified correlation for estimating HHV of solid, liquid and gaseous fuels. Fuel, 81(8), 1051-1063. https://doi.org/10.1016/S0016-2361(01)00131-4.

Reference values are transcribed from the cited publications. Figure 9 experimental markers are digitized values, not synthetic traces; their provenance and estimated uncertainty are recorded in data/validation/fig09_axial_centerline.csv. The model's 4000 K clipping, residual O2 and assumption-based radial mixing are disclosed because each affects interpretation.

End of report